

ECEN 629 Applied Convex Optimization

Chapter 2. Convex Sets

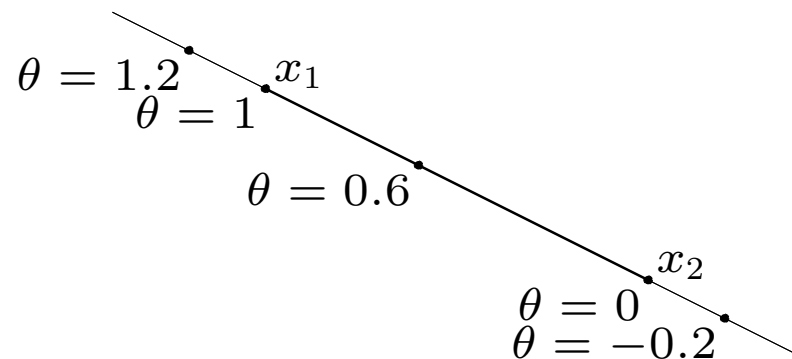
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Affine set

line through x_1, x_2 : all points

$$x = \theta x_1 + (1 - \theta)x_2 \quad (\theta \in \mathbf{R})$$



affine set: contains the line through any two distinct points in the set

example: solution set of linear equations $\{x \mid Ax = b\}$

(conversely, every affine set can be expressed as solution set of system of linear equations)

Convex set

line segment between x_1 and x_2 : all points

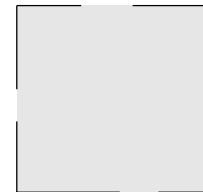
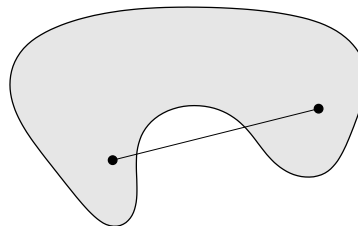
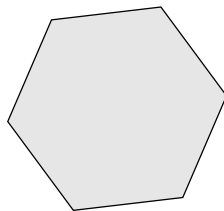
$$x = \theta x_1 + (1 - \theta)x_2$$

with $0 \leq \theta \leq 1$

convex set: contains line segment between any two points in the set

$$x_1, x_2 \in C, \quad 0 \leq \theta \leq 1 \quad \implies \quad \theta x_1 + (1 - \theta)x_2 \in C$$

examples (one convex, two nonconvex sets)



Exercise 1

Let $C \subseteq \mathbf{R}^n$ be a convex set, with $x_1, x_2, \dots, x_k \in C$. Let $\theta_1, \theta_2, \dots, \theta_k \in \mathbf{R}$ satisfies $\theta_i \geq 0$, and $\theta_1 + \theta_2 + \dots + \theta_k = 1$.

Show that $\theta_1 x_1 + \theta_2 x_2 + \dots + \theta_k x_k \in C$.

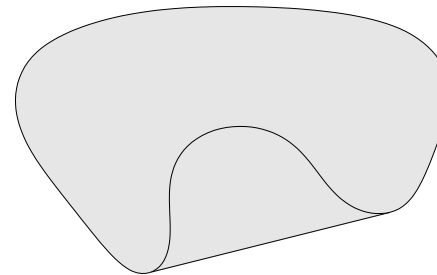
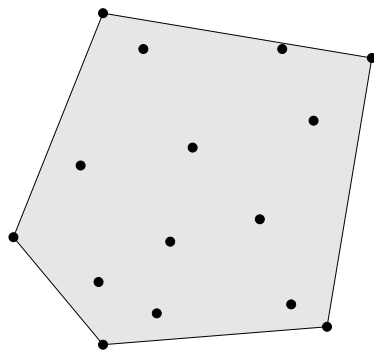
Convex combination and convex hull

convex combination of $x_1, \dots, x_k$: any point x of the form

$$x = \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_k x_k$$

with $\theta_1 + \dots + \theta_k = 1$, $\theta_i \geq 0$

convex hull $\text{conv } S$: set of all convex combinations of points in S



Exercise 2

Show that the convex hull of a set S is convex.

Show that the convex hull of a set S is the intersection of all convex sets that contains S .

Exercise 3

Determine whether the following sets are affine (convex)

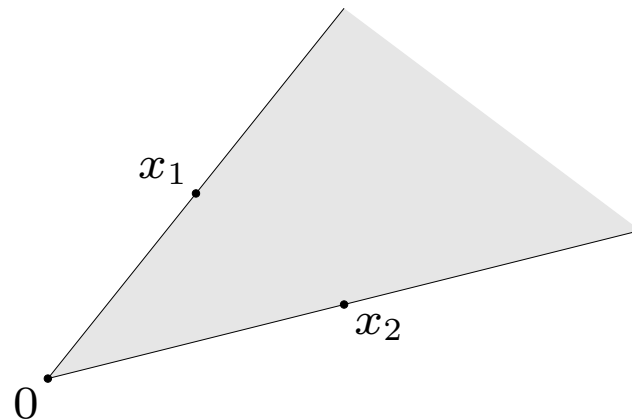
- Empty set $\emptyset$;
- A single point x_0 ;
- A line;
- A line segment;
- A ray: $\{x_0 + \theta v \mid \theta \geq 0\}$, where $v \neq 0$.

Convex cone

conic (nonnegative) combination of x_1 and x_2 : any point of the form

$$x = \theta_1 x_1 + \theta_2 x_2$$

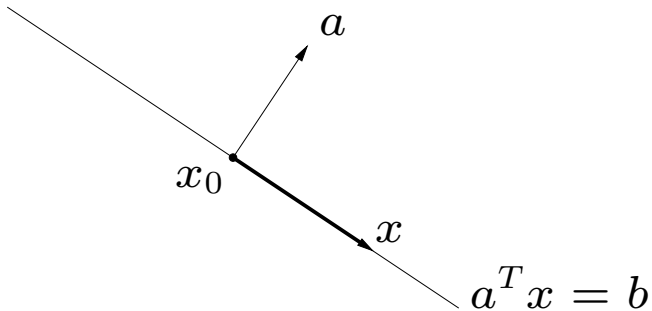
with $\theta_1 \geq 0$, $\theta_2 \geq 0$



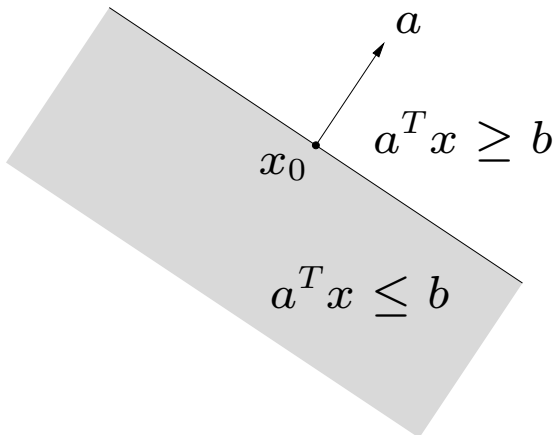
convex cone: set that contains all conic combinations of points in the set

Hyperplanes and halfspaces

hyperplane: set of the form $\{x \mid a^T x = b\}$ ($a \neq 0$)



halfspace: set of the form $\{x \mid a^T x \leq b\}$ ($a \neq 0$)



- a is the normal vector
- hyperplanes are affine and convex; halfspaces are convex

Exercise 4

When does one halfspace contain another, i.e.,

$$\{x | a^T x \leq b\} \subseteq \{x | \tilde{a}^T x \leq \tilde{b}\}?$$

Euclidean balls and ellipsoids

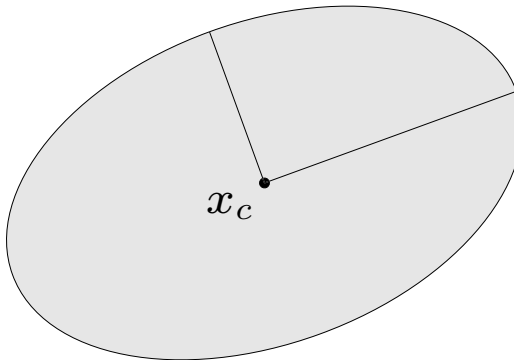
(Euclidean) ball with center x_c and radius r :

$$B(x_c, r) = \{x \mid \|x - x_c\|_2 \leq r\} = \{x_c + ru \mid \|u\|_2 \leq 1\}$$

ellipsoid: set of the form

$$\{x \mid (x - x_c)^T P^{-1} (x - x_c) \leq 1\}$$

with $P \in \mathbf{S}_{++}^n$ (*i.e.*, P symmetric positive definite)



other representation: $\{x_c + Au \mid \|u\|_2 \leq 1\}$ with A square and nonsingular

Norm balls and norm cones

norm: a function $\|\cdot\|$ that satisfies

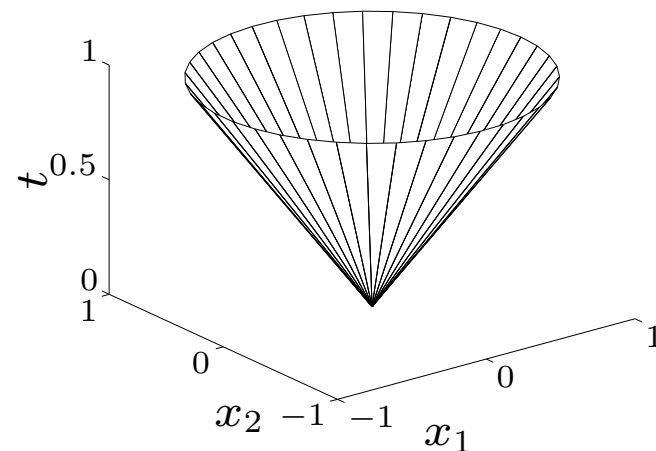
- $\|x\| \geq 0$; $\|x\| = 0$ if and only if $x = 0$
- $\|tx\| = |t| \|x\|$ for $t \in \mathbf{R}$
- $\|x + y\| \leq \|x\| + \|y\|$

notation: $\|\cdot\|$ is general (unspecified) norm; $\|\cdot\|_{\text{symb}}$ is particular norm

norm ball with center x_c and radius r : $\{x \mid \|x - x_c\| \leq r\}$

norm cone: $\{(x, t) \mid \|x\| \leq t\}$

Euclidean norm cone is called second-order cone



norm balls and cones are convex

Exercise 5

Show that any norm ball is convex.

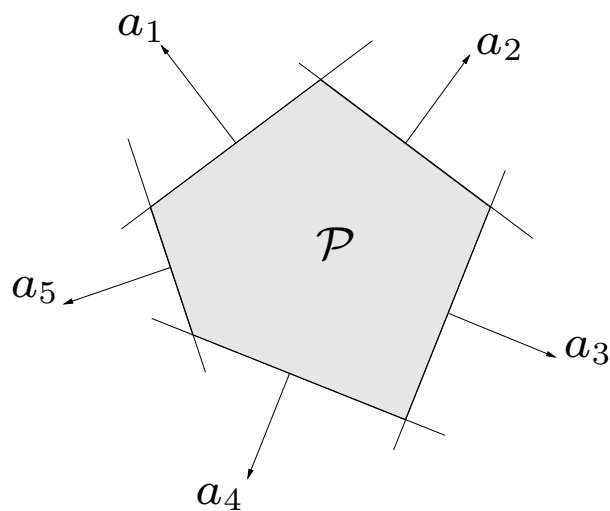
Draw the norm cone when x is one dimensional, and $\|x\| = \|x\|_2$.

Polyhedra

solution set of finitely many linear inequalities and equalities

$$Ax \preceq b, \quad Cx = d$$

($A \in \mathbf{R}^{m \times n}$, $C \in \mathbf{R}^{p \times n}$, $\preceq$ is componentwise inequality)



polyhedron is intersection of finite number of halfspaces and hyperplanes

Subtlety of representation

A polytope (bounded polyhedra) can be represented as

- Intersection of finite number of halfspaces and hyperplanes;
- A convex hull of a finite set of points.

Which is better?

Example: $x = (x_1, x_2, \dots, x_n) \in \mathbf{R}^n$

$$\{x \mid |x_i| \leq 1, i = 1, 2, \dots, n\}. \quad \text{the first representation}$$

What is the second representation?

Positive semidefinite cone

notation:

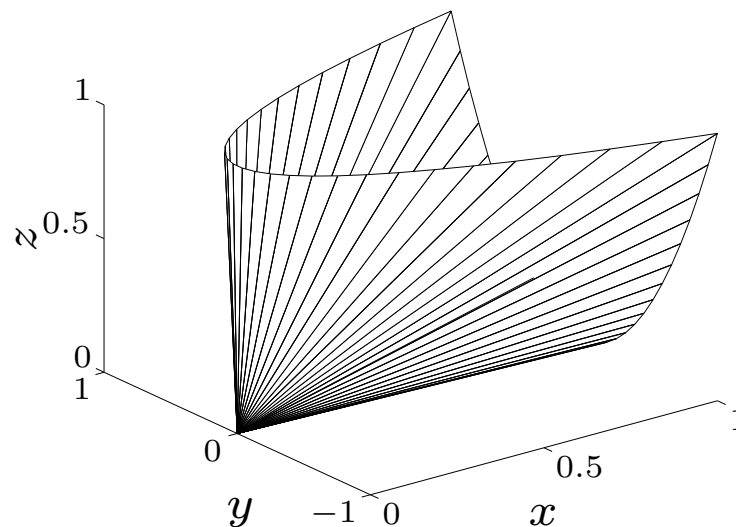
- $\mathbf{S}^n$ is set of symmetric $n \times n$ matrices
- $\mathbf{S}_+^n = \{X \in \mathbf{S}^n \mid X \succeq 0\}$: positive semidefinite $n \times n$ matrices

$$X \in \mathbf{S}_+^n \iff z^T X z \geq 0 \text{ for all } z$$

$\mathbf{S}_+^n$ is a convex cone

- $\mathbf{S}_{++}^n = \{X \in \mathbf{S}^n \mid X \succ 0\}$: positive definite $n \times n$ matrices

example: $\begin{bmatrix} x & y \\ y & z \end{bmatrix} \in \mathbf{S}_+^2$



Operations that preserve convexity

practical methods for establishing convexity of a set C

1. apply definition

$$x_1, x_2 \in C, \quad 0 \leq \theta \leq 1 \quad \implies \quad \theta x_1 + (1 - \theta)x_2 \in C$$

2. show that C is obtained from simple convex sets (hyperplanes, halfspaces, norm balls, . . .) by operations that preserve convexity

- intersection
- affine functions
- perspective function
- linear-fractional functions

Intersection

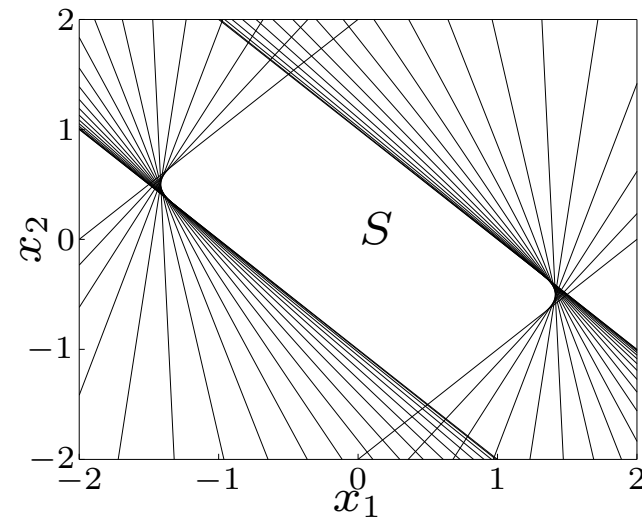
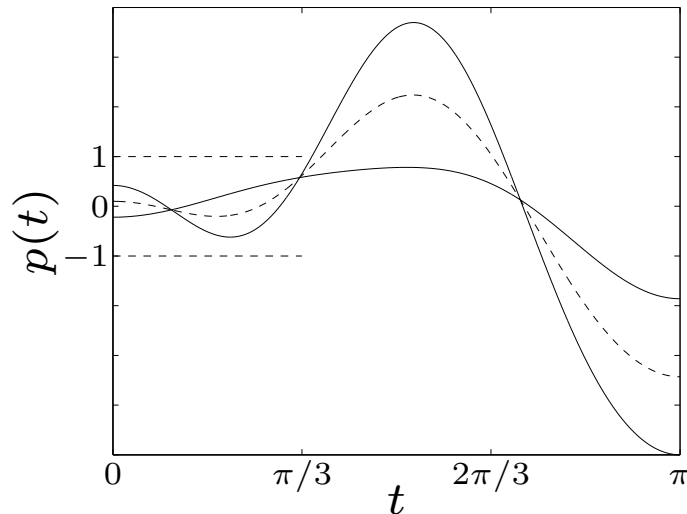
the intersection of (any number of) convex sets is convex

example:

$$S = \{x \in \mathbf{R}^m \mid |p(t)| \leq 1 \text{ for } |t| \leq \pi/3\}$$

where $p(t) = x_1 \cos t + x_2 \cos 2t + \cdots + x_m \cos mt$

for $m = 2$:



Two proofs that $\mathbf{S}_+^n$ is convex

Proof 1:

$$\begin{aligned} A, B \in \mathbf{S}_+^n &\Rightarrow \theta A + (1 - \theta)B \in \mathbf{S}_+^n \\ \forall x, x^T A x \geq 0, x^T B x \geq 0 &\Rightarrow \forall x, x^T (\theta A + (1 - \theta)B) x \geq 0 \end{aligned}$$

Proof 2:

$$\bigcap_{z \neq 0} \{X \in \mathbf{S}^n \mid z^T X z \geq 0\}.$$

Affine function

suppose $f : \mathbf{R}^n \rightarrow \mathbf{R}^m$ is affine ($f(x) = Ax + b$ with $A \in \mathbf{R}^{m \times n}$, $b \in \mathbf{R}^m$)

- the image of a convex set under f is convex

$$S \subseteq \mathbf{R}^n \text{ convex} \implies f(S) = \{f(x) \mid x \in S\} \text{ convex}$$

- the inverse image $f^{-1}(C)$ of a convex set under f is convex

$$C \subseteq \mathbf{R}^m \text{ convex} \implies f^{-1}(C) = \{x \in \mathbf{R}^n \mid f(x) \in C\} \text{ convex}$$

examples

- scaling, translation, projection
- solution set of linear matrix inequality $\{x \mid x_1 A_1 + \cdots + x_m A_m \preceq B\}$
(with $A_i, B \in \mathbf{S}^p$)
- hyperbolic cone $\{x \mid x^T P x \leq (c^T x)^2, c^T x \geq 0\}$ (with $P \in \mathbf{S}_+^n$)

Mapping with Affine Functions

Scaling, translation and projection

- Scaling: $\alpha S \triangleq \{\alpha x | x \in S\};$
- Translation: $S + a \triangleq \{x + a | x \in S\};$
- Projection on some coordinates:
 $T \triangleq \{x_1 \in \mathbf{R}^m | (x_1, x_2) \in S \text{ for some } x_2 \in \mathbf{R}^n\}.$

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Polyhedron: $C = \{x | Ax \succeq b, Cx = d\}$

- Affine function: $f(x) = (b - Ax, d - Cx);$
- The inverse image of the Cartesian product of the nonnegative orthant and the origin

$$C = \{x | f(x) \in \mathbf{R}_+^m \times \{0\}\}.$$

Mapping with Affine Functions

Linear matrix inequality: $\{x \mid \sum_{i=1}^m x_i A_i \preceq B\}$

- Affine function: $f(x) = B - A(x)$.

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Hyperbolic cone: $\{x \mid x^T P x \leq (c^T x)^2, c^T x \geq 0\}$ where $P \in \mathbf{S}_+^n$

- Affine function: $f(x) = (P^{1/2} x, c^T x)$;

Mapping with Affine Functions

Linear matrix inequality: $\{x \mid \sum_{i=1}^m x_i A_i \preceq B\}$

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- Affine function: $f(x) = (P^{1/2} x, c^T x)$;
- Second order cone: $\{(z, t) \mid z^T z \leq t^2, t \geq 0\}$.

Perspective and linear-fractional function

perspective function $P : \mathbf{R}^{n+1} \rightarrow \mathbf{R}^n$:

$$P(x, t) = x/t, \quad \text{dom } P = \{(x, t) \mid t > 0\}$$

images and inverse images of convex sets under perspective are convex

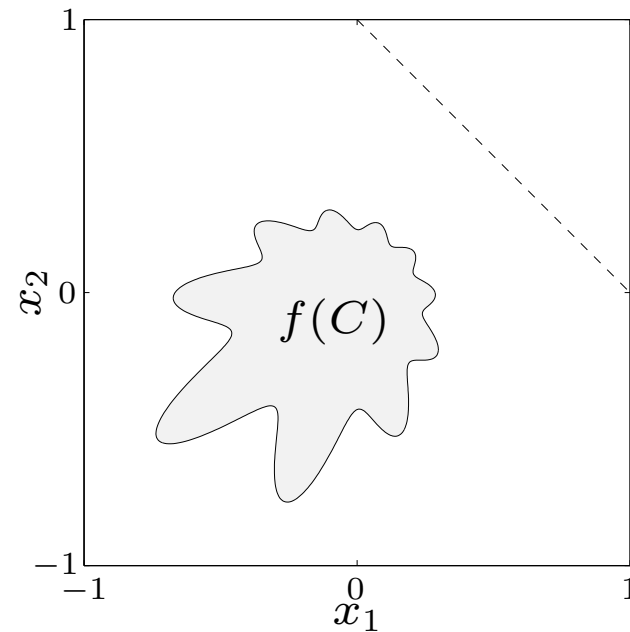
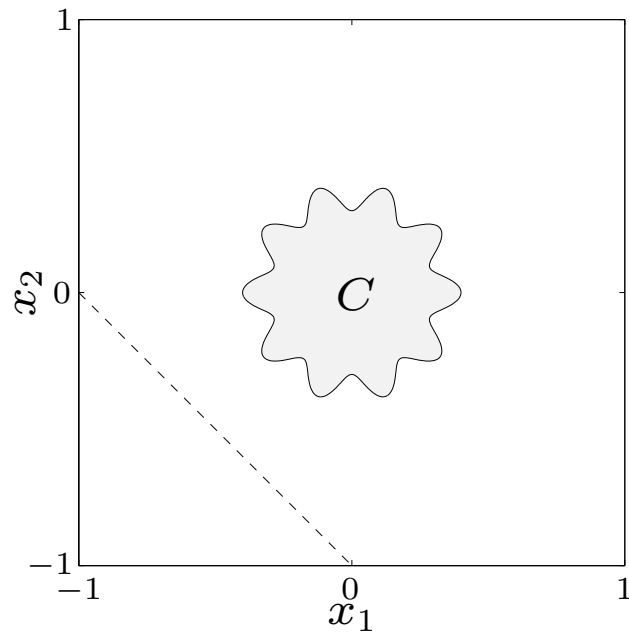
linear-fractional function $f : \mathbf{R}^n \rightarrow \mathbf{R}^m$:

$$f(x) = \frac{Ax + b}{c^T x + d}, \quad \text{dom } f = \{x \mid c^T x + d > 0\}$$

images and inverse images of convex sets under linear-fractional functions are convex

example of a linear-fractional function

$$f(x) = \frac{1}{x_1 + x_2 + 1}x$$



Proof: $P(x)$ Preserves Convexity

Forward: consider $x = (\tilde{x}, x_{n+1}) \in C$ and $y = (\tilde{y}, y_{n+1}) \in C$, we'll prove any line segment is mapped to a line segment.

- Need to prove $P(\theta x + (1 - \theta)y) \in P(C)$;

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$$P(\theta x + (1 - \theta)y) = \frac{\theta \tilde{x} + (1 - \theta)\tilde{y}}{\theta x_{n+1} + (1 - \theta)y_{n+1}}$$

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- Define

$$\mu \triangleq \frac{\theta x_{n+1}}{\theta x_{n+1} + (1 - \theta)y_{n+1}}$$

In fact, for any $\theta \in [0, 1]$, there is a corresponding $\mu \in [0, 1]$, and vice versa;

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In fact, for any $\theta \in [0, 1]$, there is a corresponding $\mu \in [0, 1]$, and vice versa;

- $P(\theta x + (1 - \theta)y) = \mu P(x) + (1 - \mu)P(y)$.

Now suppose the convex set $C \subseteq \text{dom}(P)$, and $P(x), P(y) \in P(C)$, then the whole line segment between $P(x)$ and $P(y)$ are in $P(C)$.

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Forward: consider $x = (\tilde{x}, x_{n+1}) \in C$ and $y = (\tilde{y}, y_{n+1}) \in C$, we'll prove any line segment is mapped to a line segment.

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- $P(\theta x + (1 - \theta)y) = \mu P(x) + (1 - \mu)P(y)$.

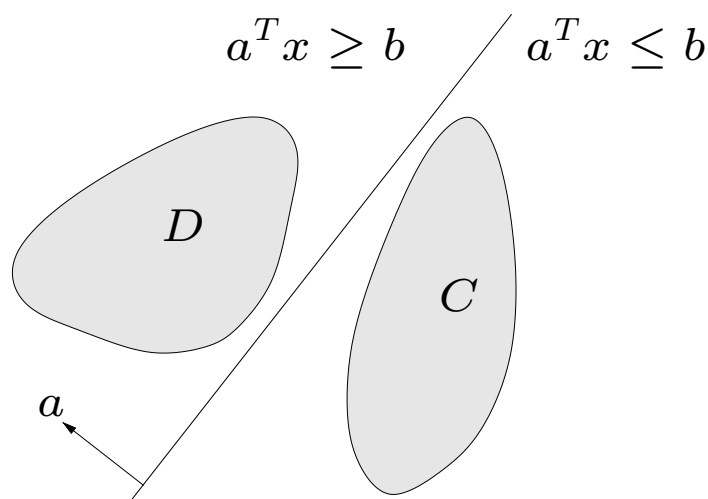
Now suppose the convex set $C \subseteq \mathbf{dom}(P)$, and $P(x), P(y) \in P(C)$, then the whole line segment between $P(x)$ and $P(y)$ are in $P(C)$.

Inverse image: requires another (similar) proof.

Separating hyperplane theorem

if C and D are nonempty disjoint convex sets, there exist $a \neq 0$, b s.t.

$$a^T x \leq b \text{ for } x \in C, \quad a^T x \geq b \text{ for } x \in D$$



the hyperplane $\{x \mid a^T x = b\}$ separates C and D

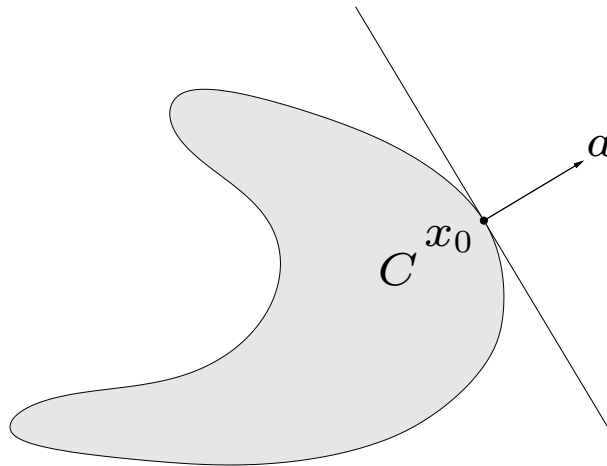
strict separation requires additional assumptions (*e.g.*, C is closed, D is a singleton)

Supporting hyperplane theorem

supporting hyperplane to set C at boundary point x_0 :

$$\{x \mid a^T x = a^T x_0\}$$

where $a \neq 0$ and $a^T x \leq a^T x_0$ for all $x \in C$



supporting hyperplane theorem: if C is convex, then there exists a supporting hyperplane at every boundary point of C

Separation of an affine and a convex set

C is convex, and $D = \{Fu + g \mid u \in \mathbf{R}^m\}$ is affine, where $F \in \mathbf{R}^{n \times m}$.

- C and D are disjoint: $a^T x \leq b$ for $x \in C$, and $a^T x \geq b$ for all $x \in D$;
- $a^T Fu \geq b - a^T g$ for all $u \in \mathbf{R}^m \Rightarrow a^T F = 0$.

Separating Hyperplane

Q: Why convex set?

- Give an example of non-separation, when one of the sets is not convex.

Separating Hyperplane

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Strict separation and characterization by intersection of hyperplanes

- Strict separation is not always possible for two convex set;
- Holds when one set is singleton and the other is closed;
- Corollary: A closed convex set is the intersection of all halfspaces that contain it.

Exercise 6

Let a and b be two distinct points in $\mathbf{R}^n$.

- Show that the set $C = \{x \mid \|x - a\|_2 \leq \|x - b\|_2\}$ is a halfspace by finding an explicit inequality of the form $c^T x \leq d$.
- What is the geometric interpretation of the set C ?
- What is the geometric interpretation of the fact that C is a halfspace.

Exercise 7

Let $p = (p_1, p_2, \dots, p_n)$ denote a probability mass function for a real-valued random variable x . Fix values $x = (a_1, a_2, \dots, a_n)$. Which set is convex?

- $\{p | \alpha \leq \mathbf{E}f(x) \leq \beta\}$, where the function $f(\cdot)$ is given.
- $\{p | \mathbf{E}|x^3| \leq \alpha \mathbf{E}|x|\}$.
- $\{p | \mathbf{var}(x) \leq \alpha\}$, where $\mathbf{var}(x) = \mathbf{E}(x - \mathbf{E}x)^2$ is the variance of x .

Exercise 8

Another operation that preserves convexity: Let S_1 and S_2 be two convex sets in $\mathbf{R}^{m+n}$. Show that the partial sum

$$S = \{(x, y_1 + y_2) \mid x \in \mathbf{R}^m, y_1, y_2 \in \mathbf{R}^n, (x, y_1) \in S_1, (x, y_2) \in S_2\}$$

is also convex.

Exercise 9

Let $P(x, t) = x/t$ with domain $\mathbf{R}^n \times \mathbf{R}_{++}$. Let

$$C = \mathbf{conv}\{(v_1, t_1), (v_2, t_2), \dots, (v_K, t_K)\}$$

where $v_i \in \mathbf{R}^n$ and $t_i > 0$. What is $P(C)$?

Hint:

- Line segment maps to line segment for perspective function.